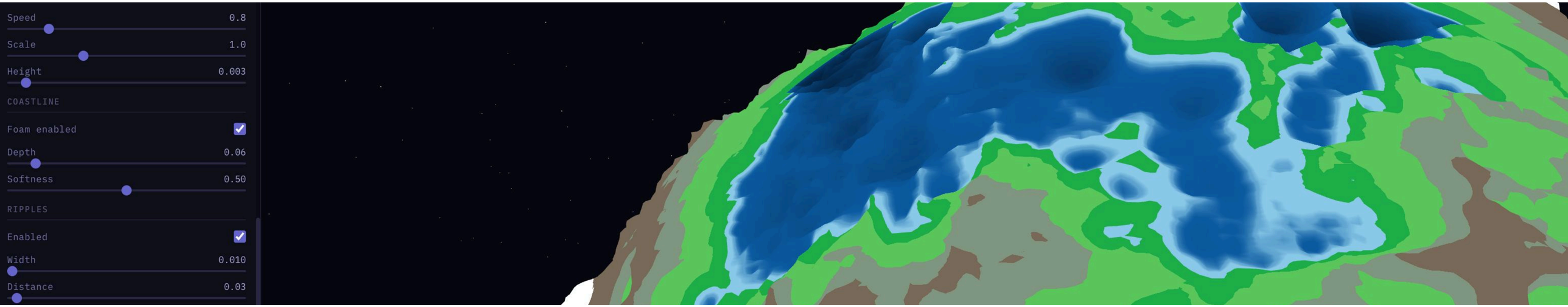
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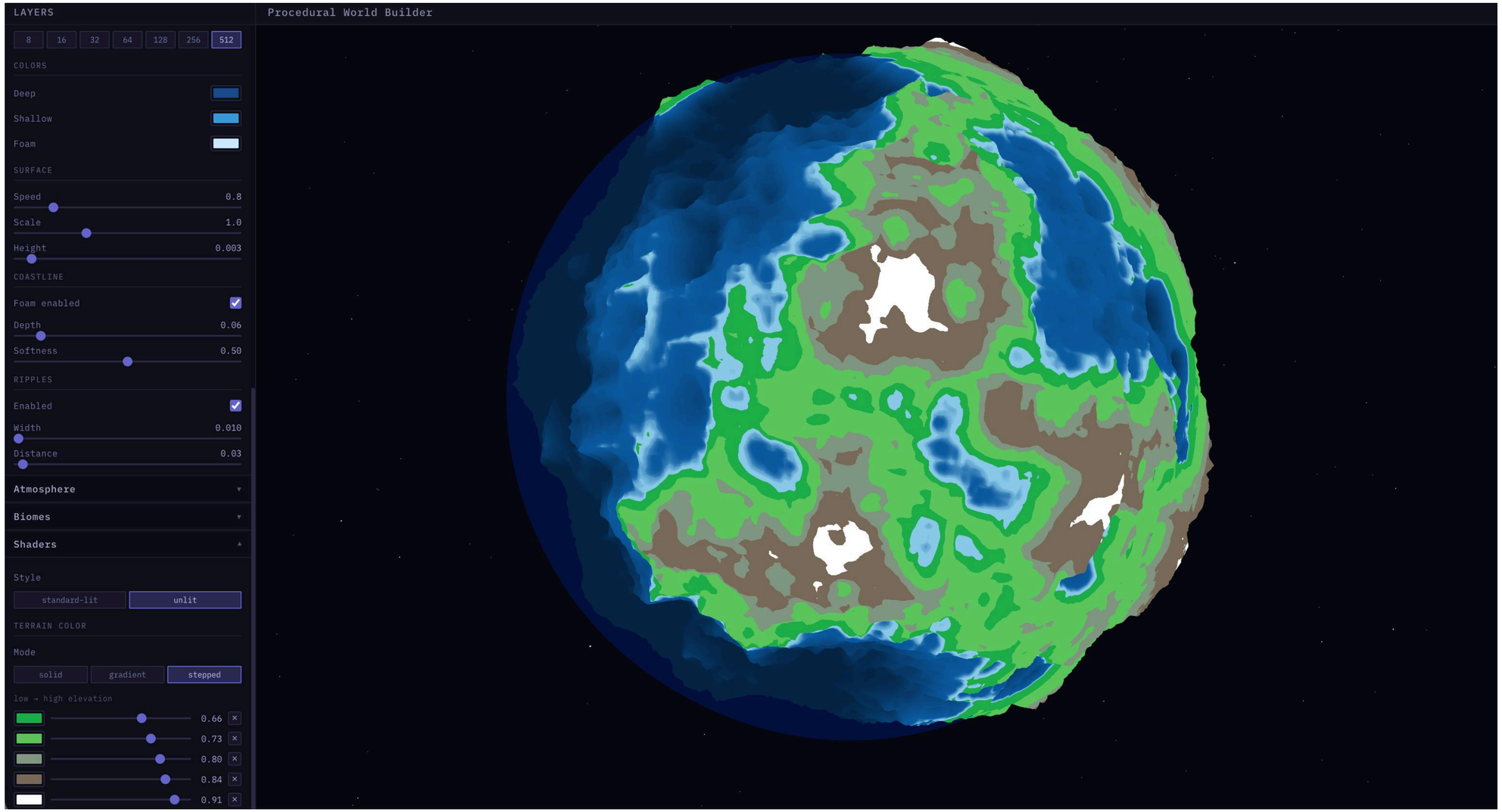
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React

three.js

Tools:
Web Frameworks
React + Three.js



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Procedural Generation Techniques

PCG 01: Maps

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01 - Pixel Array



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01 - Pixel Array

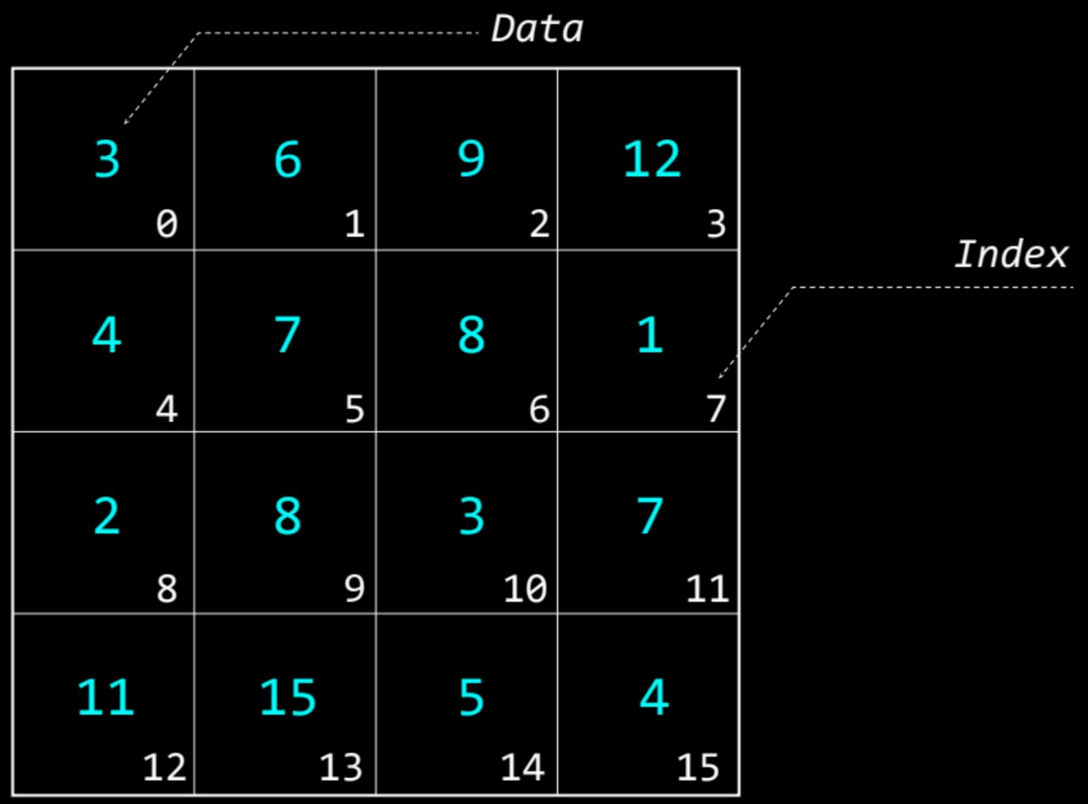
PIXEL LIST



`pixels[n]`

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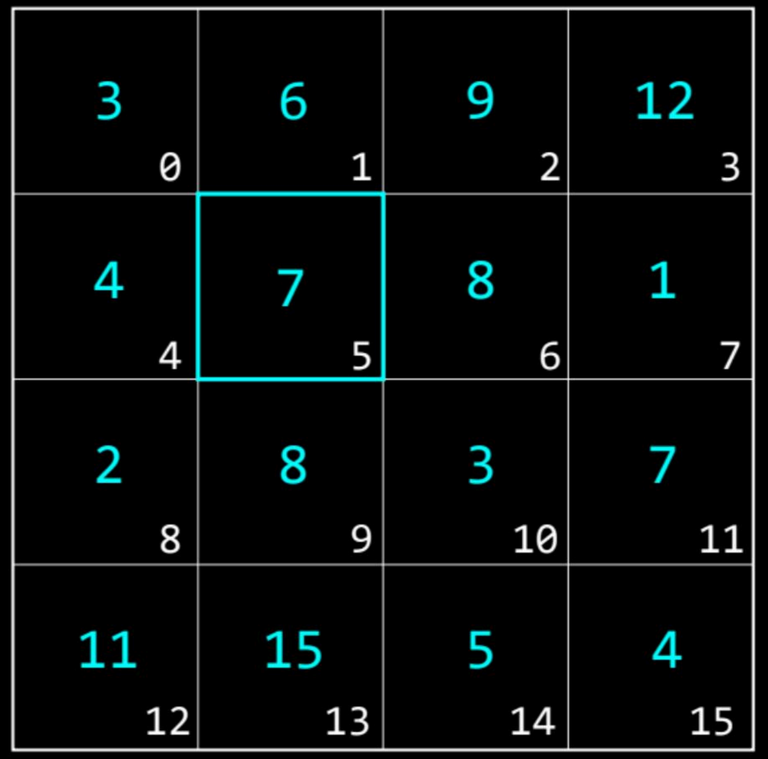
01 - Pixel Array



Index numbers of a list in a grid

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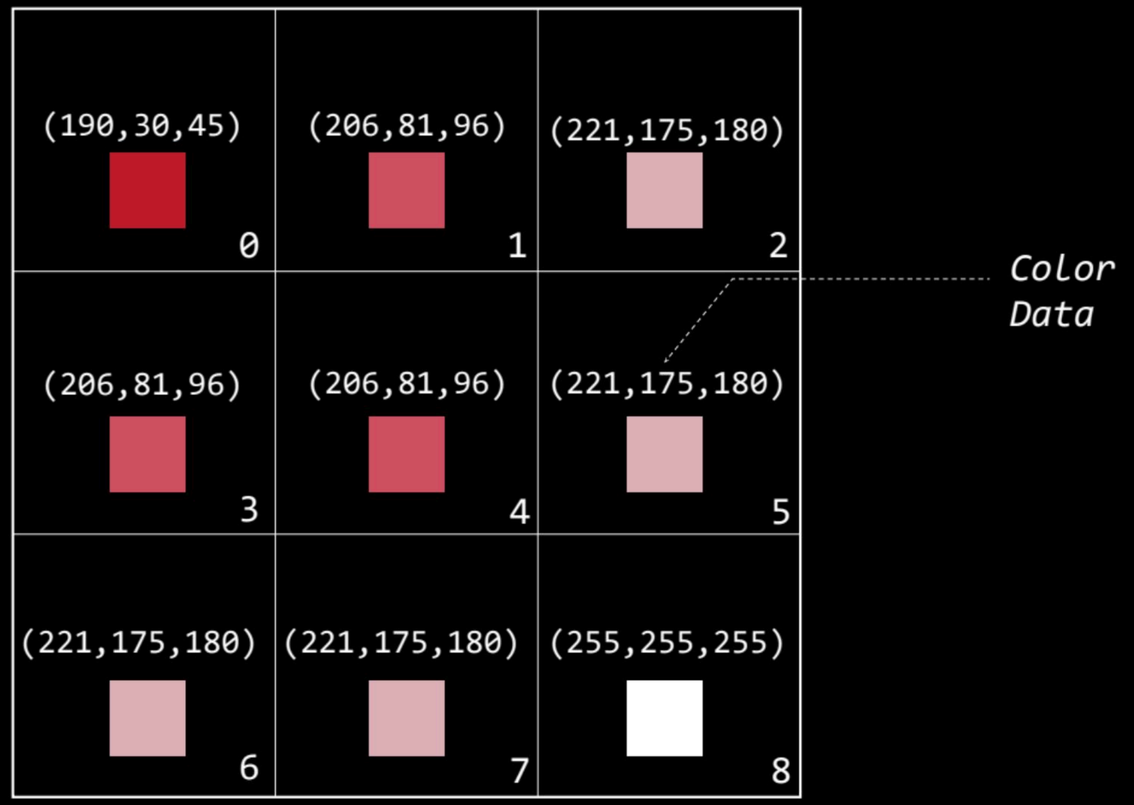
01 - Pixel Array



Index numbers of a list in a grid

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01 - Pixel Array



Grids for storing color data

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Creating Procedural Maps:

-Approach 1 - Functions

-Approach 2 - Simulation

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Approach 1:
Noise: maps as pure functions of position

Random access. You can ask for the height at any point without computing anything else. This is what makes infinite worlds possible: No Man's Sky does not store a galaxy, it evaluates one.

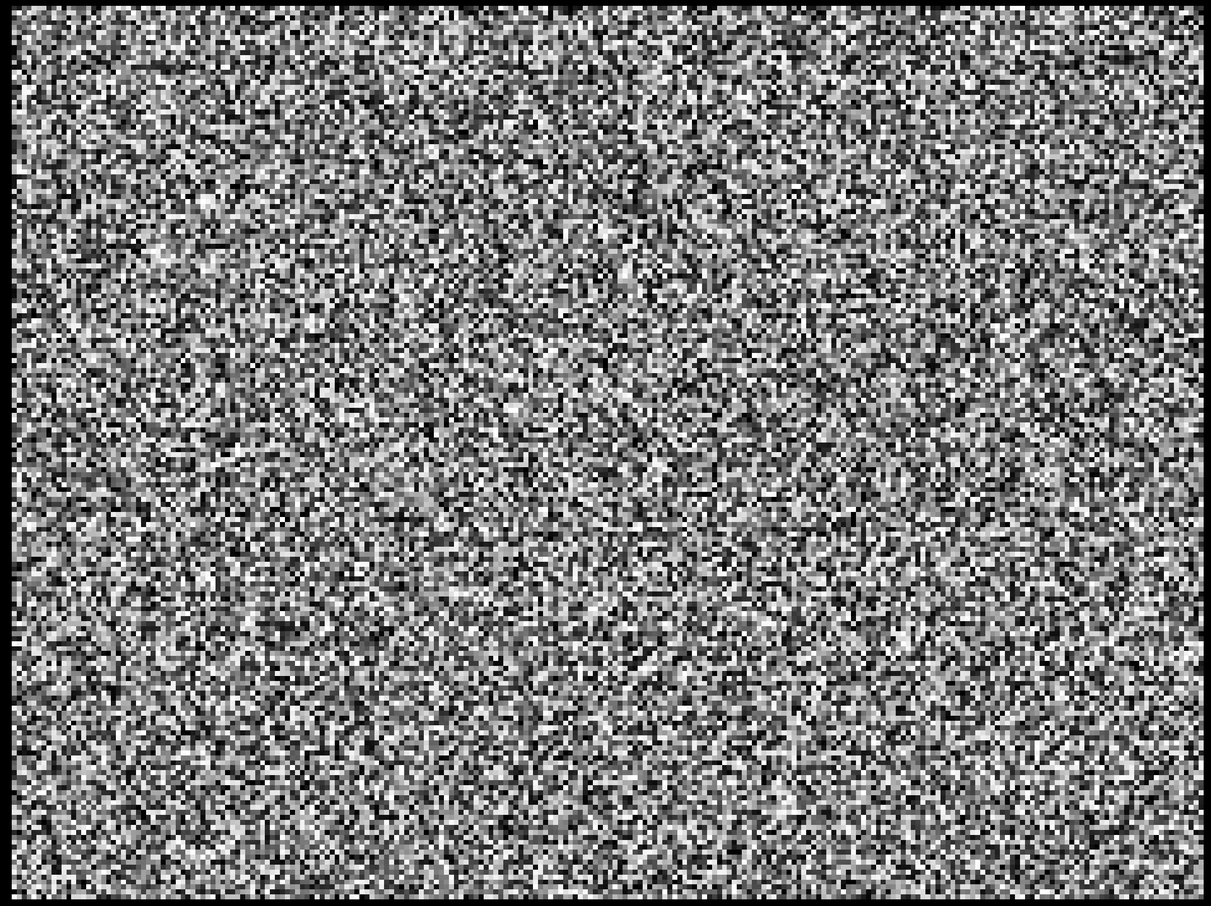
Trivially parallel. Every pixel is independent, so it maps perfectly onto GPU threads, worker threads, or a vectorised NumPy call.

Deterministic and reproducible. Same seed, same coordinates, same result — forever, on any machine, in any order. Seeds become the save file.

Chunkable. Neighbouring chunks agree automatically at their shared border, because both are evaluating the same underlying function.

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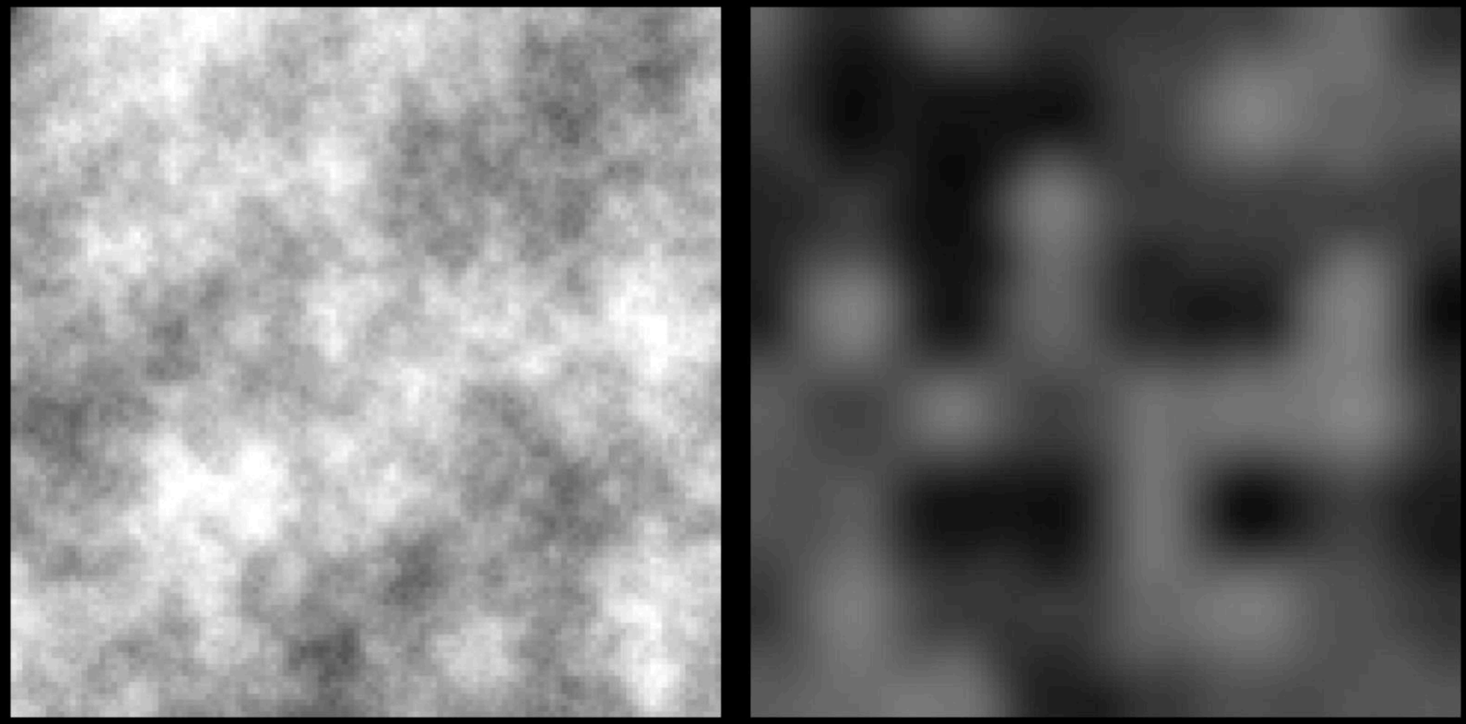
Noise equations 噪声方程



White noise - pure randomness

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Perlin Noise



Perlin Noise field

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Perlin Noise

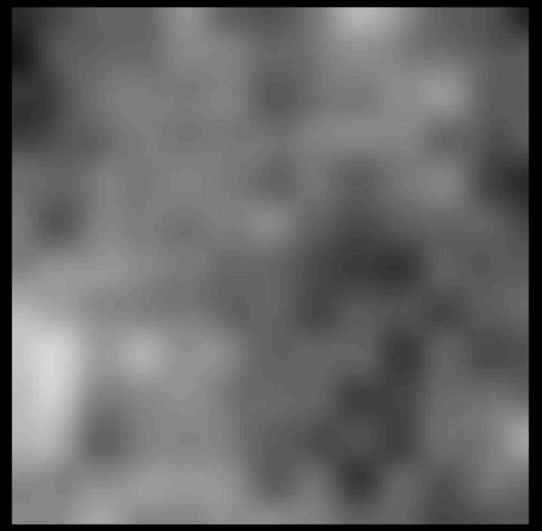
Calibration of the resulting noise field.

```
#Example

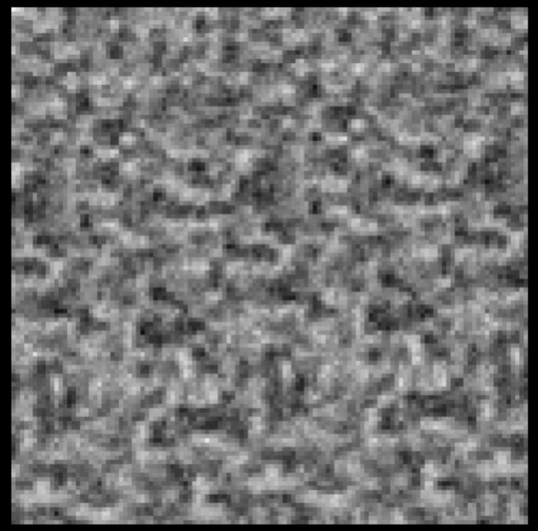
lod = 3
falloff = 0.5
noise_scale = 0.005

noiseDetail(lod, falloff)

noise_value =
noise (20 * noise_scale , 20 * noise_scale )
```



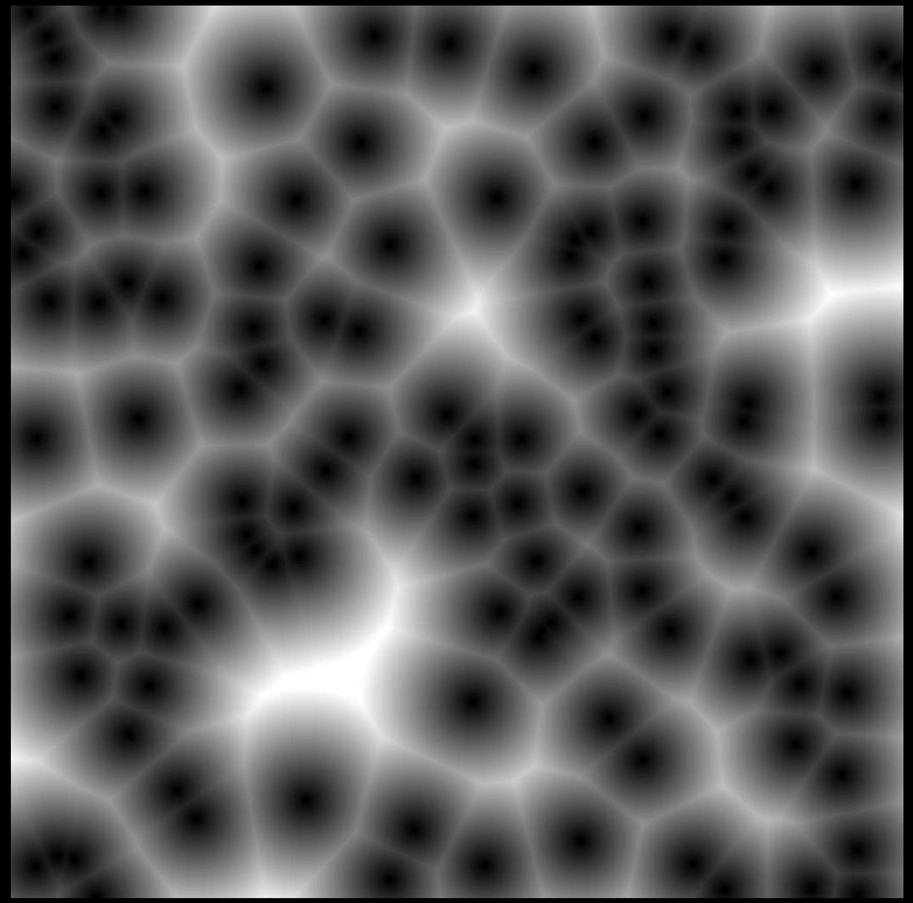
noise_scale = 0.005
lod = 3
falloff = 0.6



noise_scale = 0.06
lod = 3
falloff = 0.6

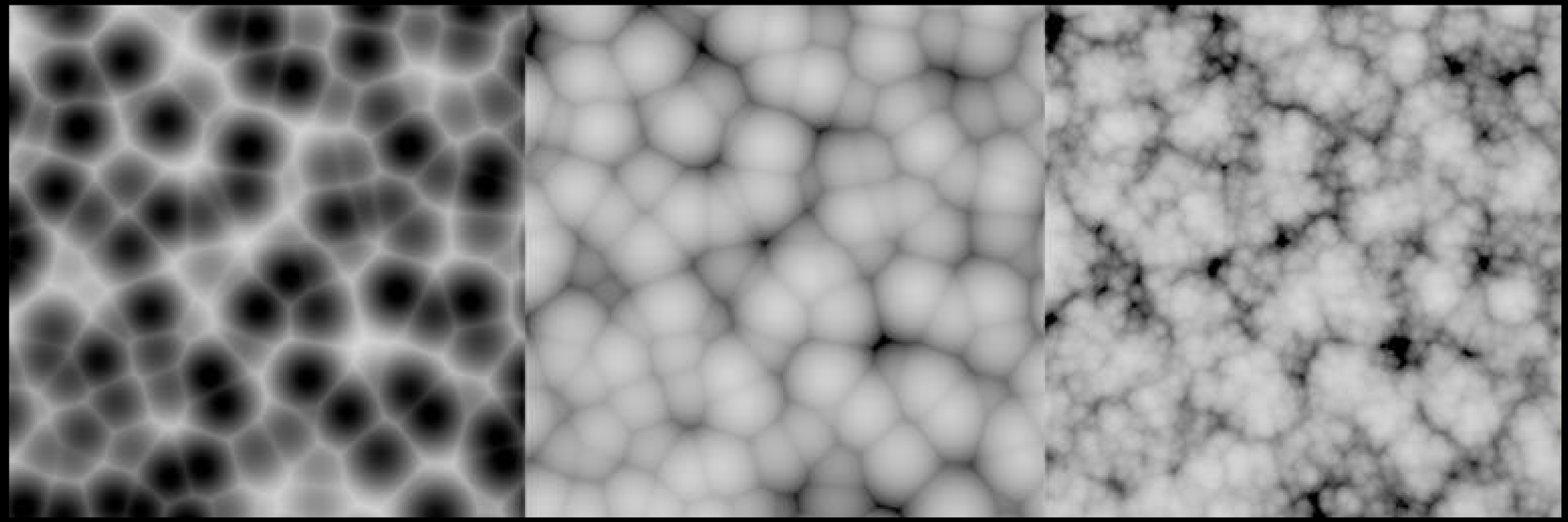
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Cellular Noise



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Cellular Noise

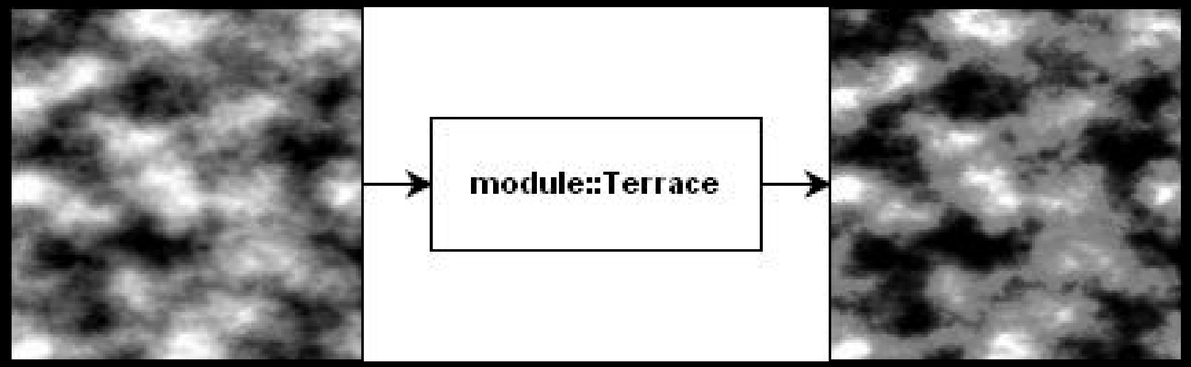


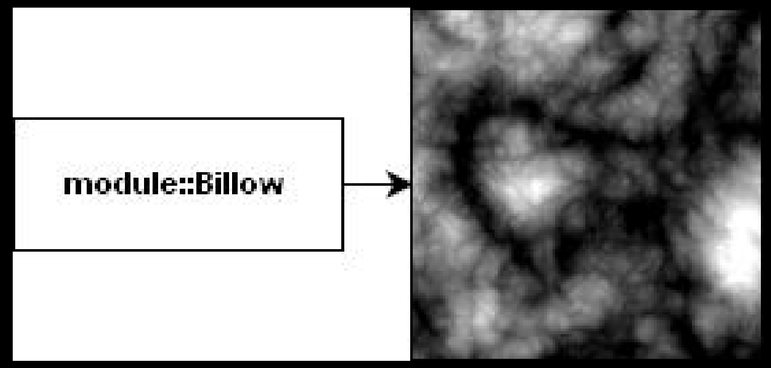
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Shaping the output

The same noise stack produces wildly different maps depending on what you do with the value afterwards. These are essentially free, and they are where most of the visual character comes from.

Consider this math modifiers for the noise values:
-Ridged
-Billow
-Turbulence
-Terracing
-Power curve
-Domain warping





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Assignment 1:

-Create a grid in your 3D view port
-Create a 2D view of a noise equation
-Apply the noise map to the 3D grid
-Include Slider controls for fine tuning the noise equation and the resolution of the grid
-Include shaping operations as a dropdown with their own slider

Ideas:
-Create layers of noise (each one with parametric control)
-Create a dropdown menu to select what noise equation to use
-Create a layer system that can allow you to see the ‘blend’ between different noise equations

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Approach 2:
Noise maps as the result of **simulations**

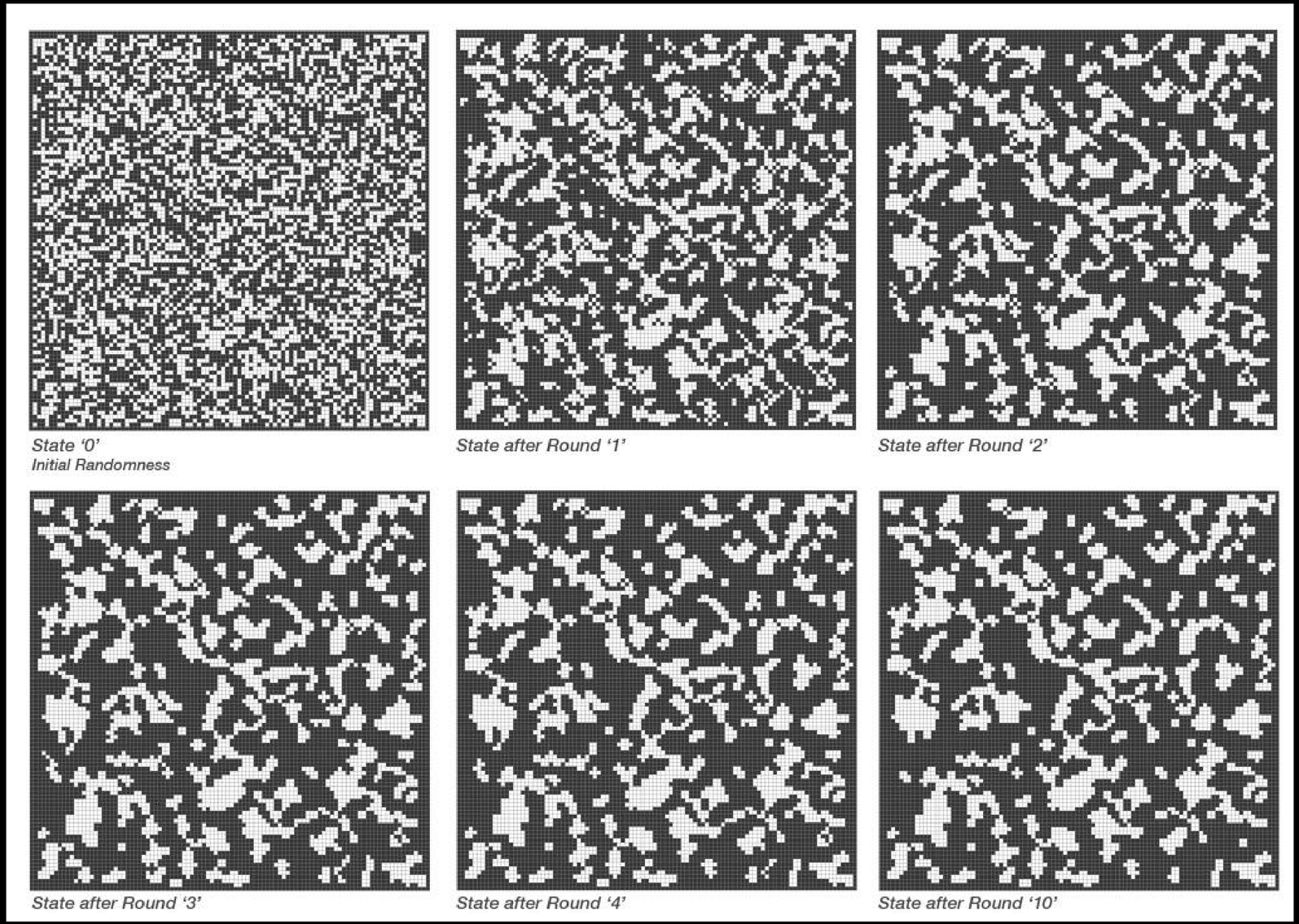
Everything so far has been a pure function.

Now we switch paradigms entirely.

Simulation methods hold the whole grid in memory and repeatedly update it based on local neighbourhoods. Structure emerges over time rather than being evaluated at a point.

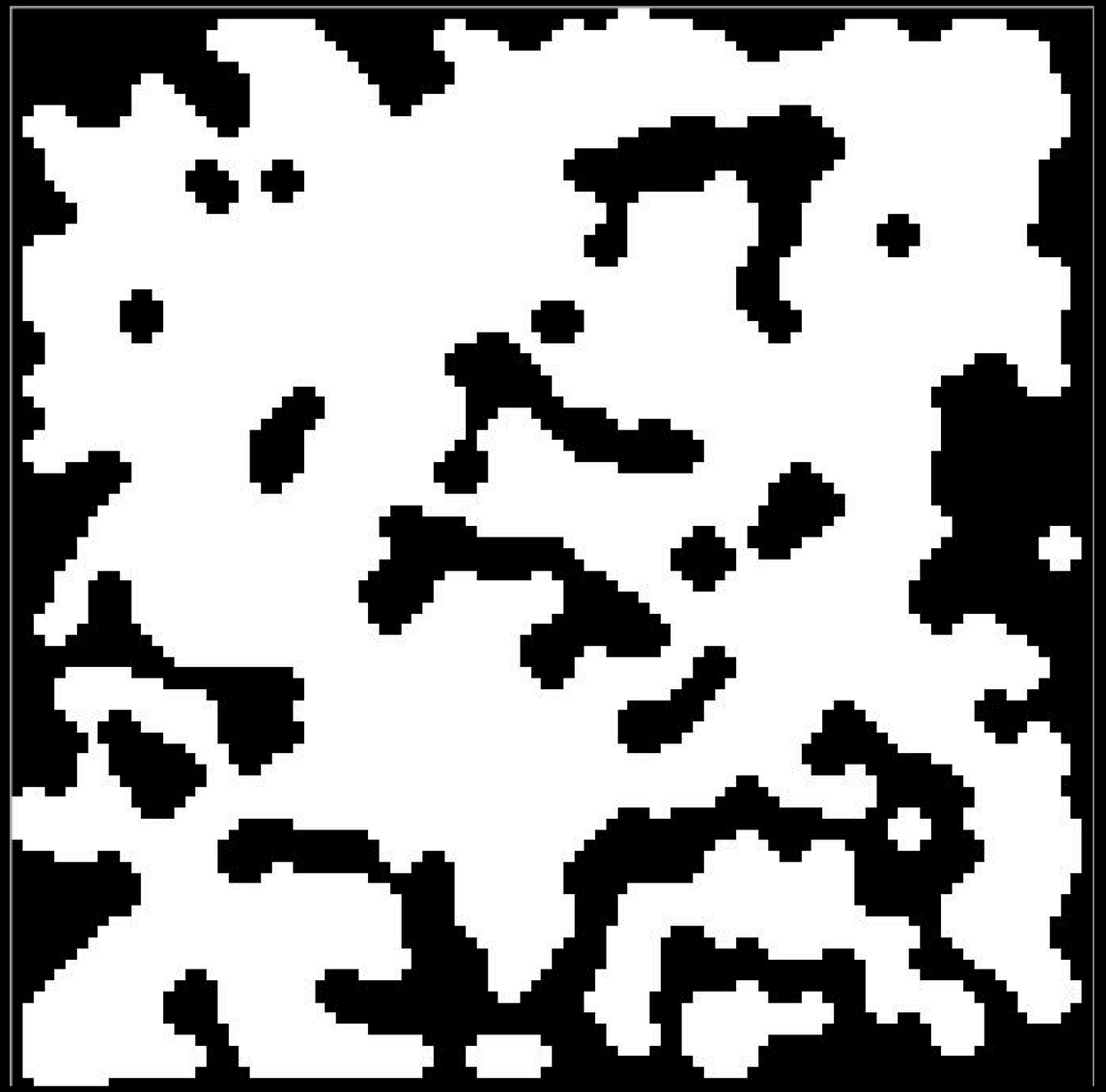
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Cellular Automata



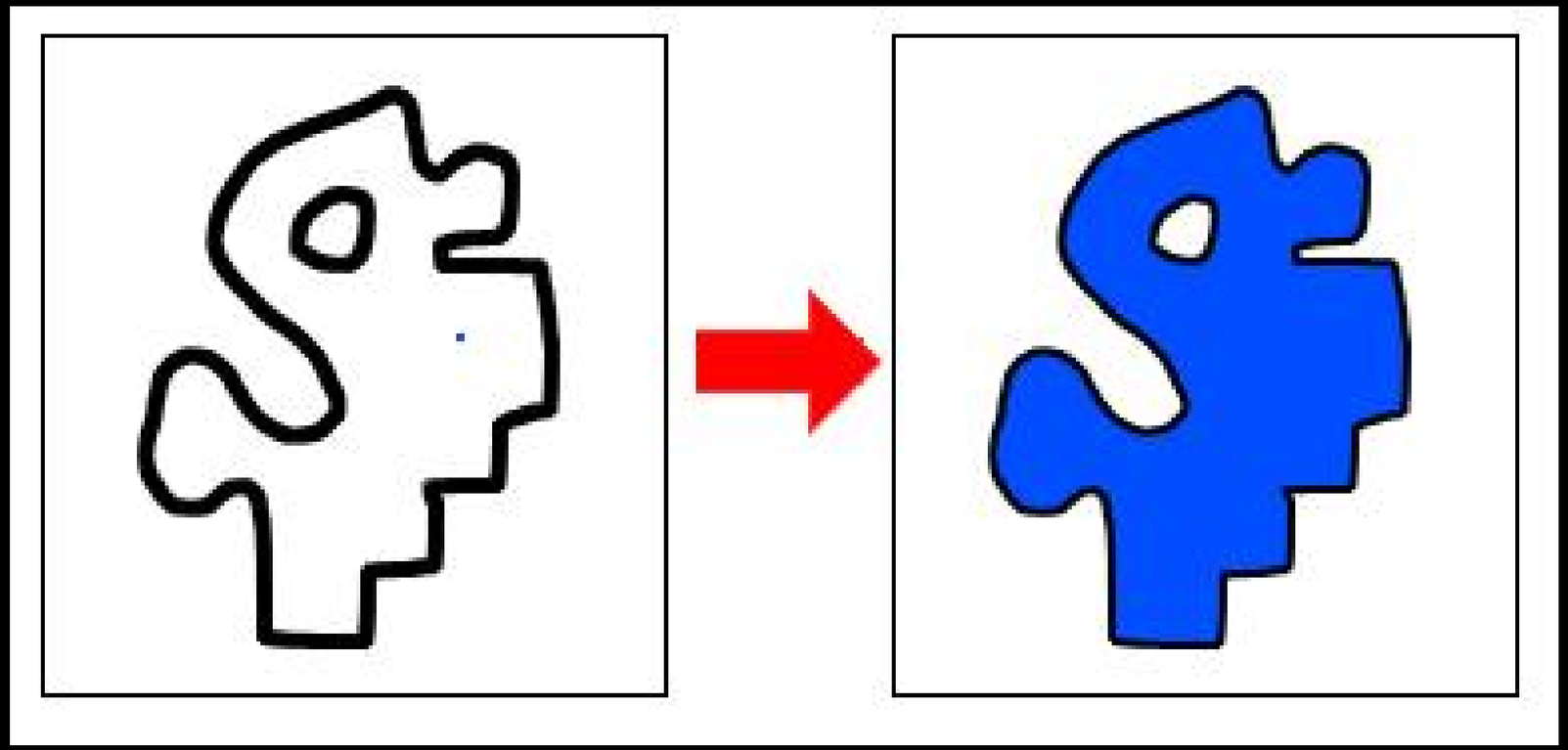
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Cellular Automata



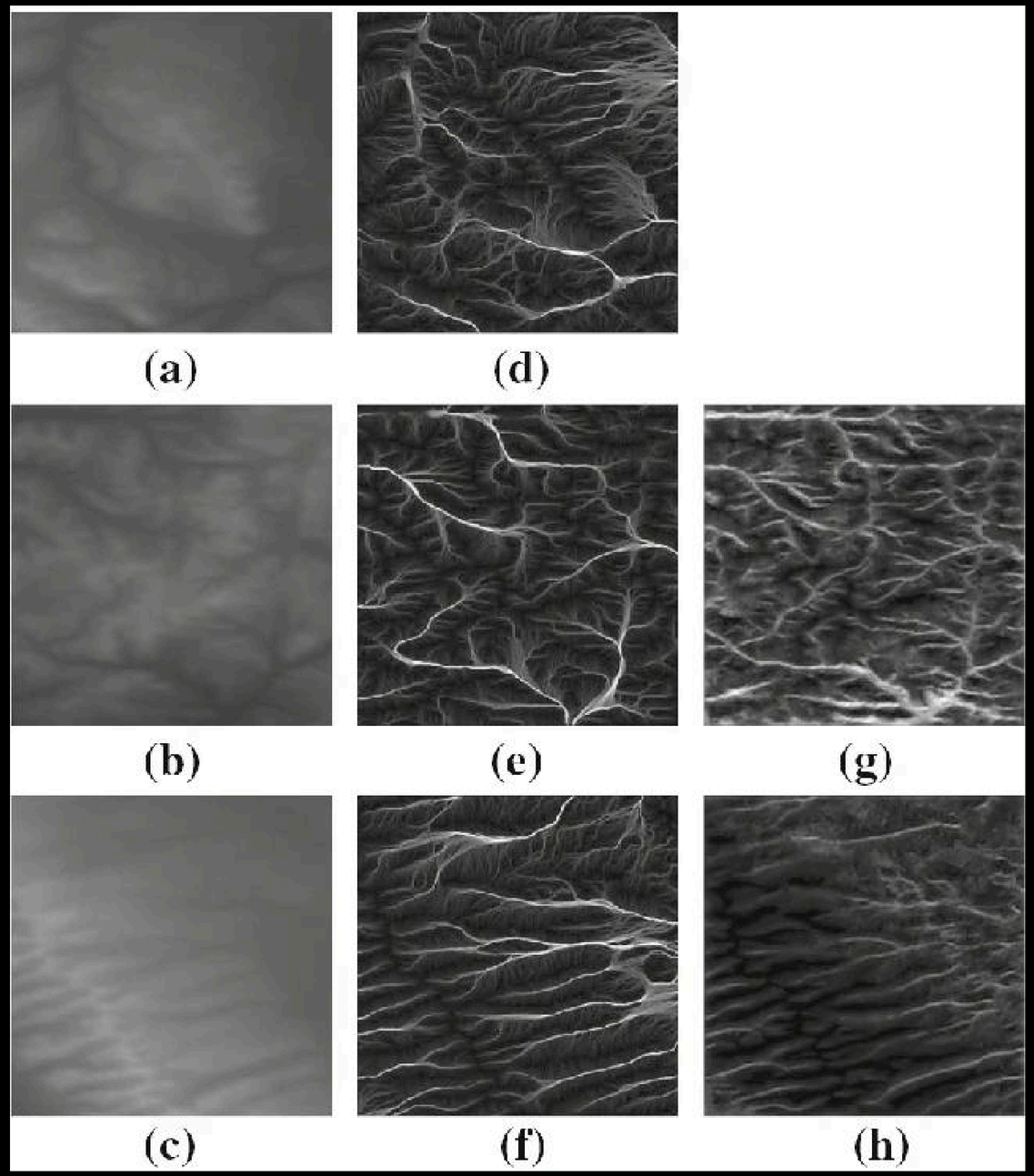
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Flood fill



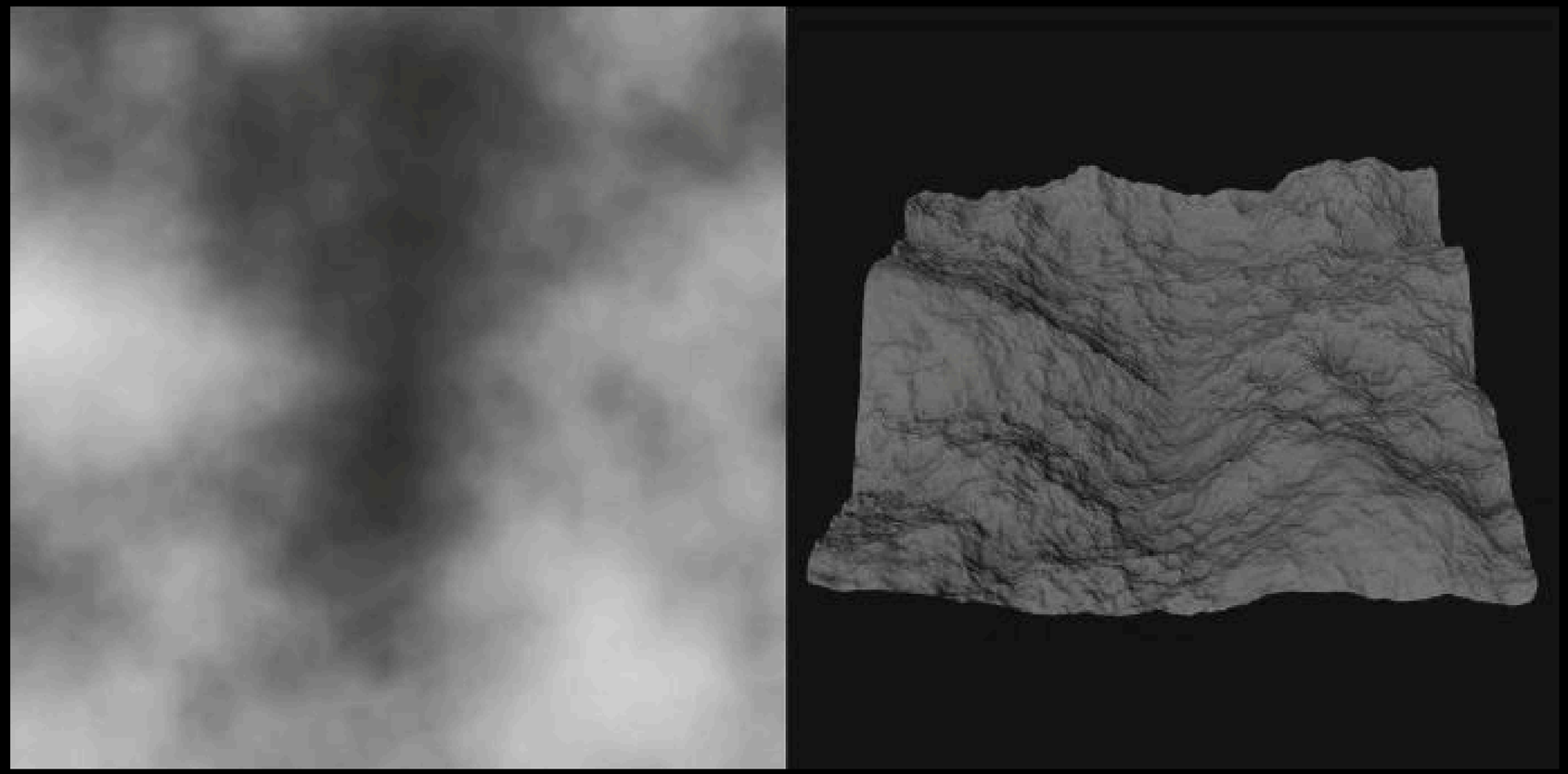
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Hydraulic Erosion Simulations



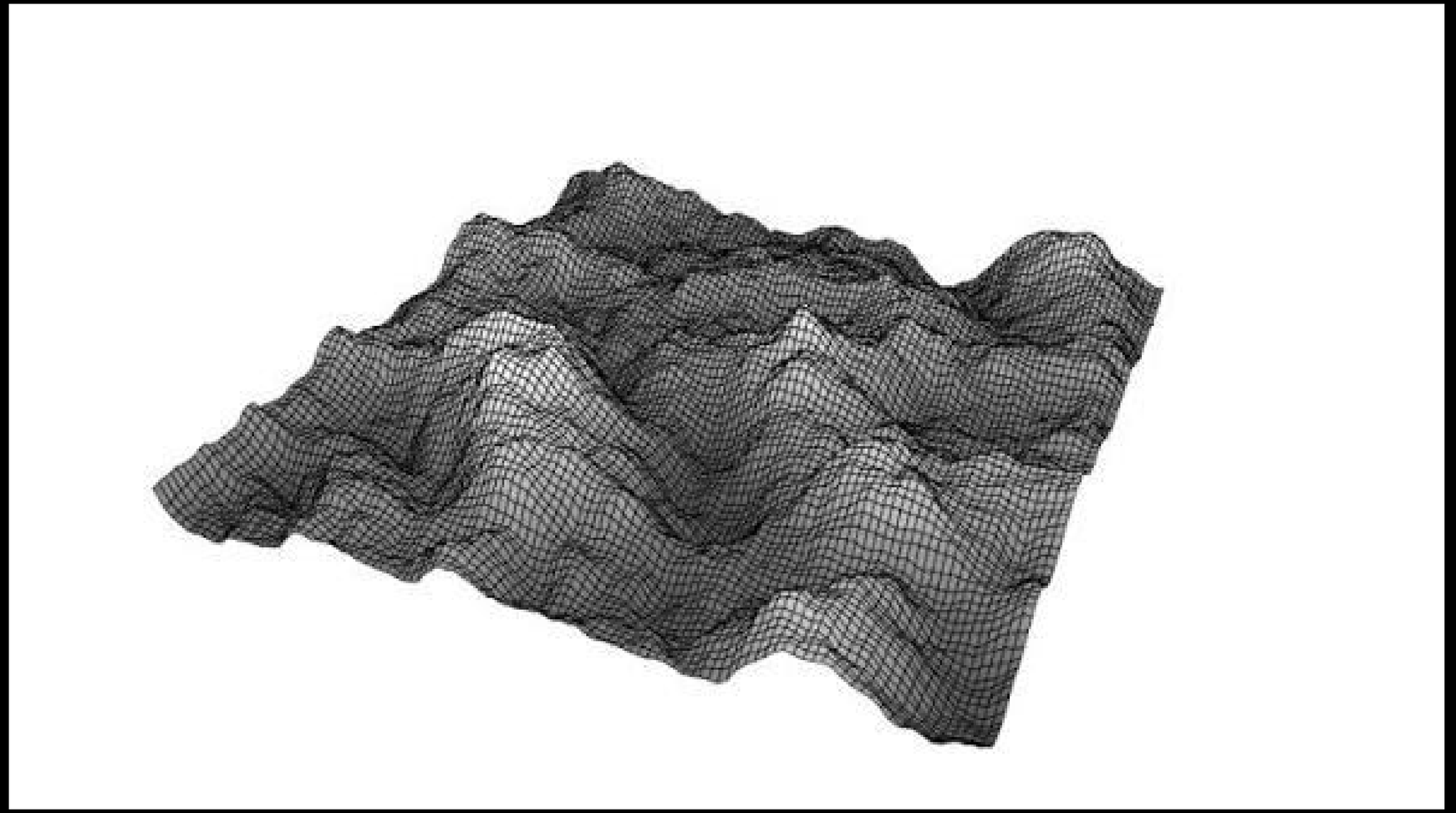
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Creating height fields



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Creating height fields



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Assignment 2:

-Add a simulation map section to your app, that starts and stops the simulation
-Create a height field that is driven by your noise stack
-Calibrate your maps and terrain resolution to achieve a believable topography

Ideas:
-Create a material or “shader” that changes based on the height
-Create an “infinite map scroller” that allows to use the keyboard to navigate the map
-Create a shortcut to toggle on/off wireframe